Proposition 4.2: A Shorter Proof

Brieuc Lehmann

29 October 2026

1 Statement and notation

Proposition 4.2 holds, and its proof takes three short steps. This note replaces the eight-step proof in Section 4.3; the statement is unchanged.

The two data sources $k = 1, 2$ are independent, and each is split into a training fold and an evaluation fold. Everything in the proof is computed on the evaluation fold, so the "eval" subscript is dropped: n_k is the size of source k's evaluation fold, $\hat{L}_k$ its lower-bound estimate and $\hat{\sigma}_k^2$ its variance estimate. Each source gives a valid identified band, so $L_k \le \ell$ for $k = 1, 2$, where ℓ is the lower end of the true band (L_{true} in the report). Write $z = \Phi^{-1}(1 - \alpha/2)$ and $s_k = \sigma_k^2/n_k$. For a weight $\omega \in [0, 1]$:

$$\hat{L}(\omega) = \omega\,\hat{L}_1 + (1-\omega)\,\hat{L}_2, \qquad L(\omega) = \omega L_1 + (1-\omega)L_2,$$
$$v_n(\omega) = \omega^2 s_1 + (1-\omega)^2 s_2, \qquad \hat{v}_n(\omega) = \omega^2 \frac{\hat{\sigma}_1^2}{n_1} + (1-\omega)^2 \frac{\hat{\sigma}_2^2}{n_2},$$
$$\hat{A}(\omega) = \hat{L}(\omega) - z\sqrt{\hat{v}_n(\omega)}.$$

Sample sizes are indexed by n, with $n_1, n_2 \to \infty$. Assumptions:

(A1) The training and evaluation folds are independent.

(A2) For $k = 1, 2$, $\sqrt{n_k}\,(\hat{L}_k - L_k) \xrightarrow{d} N(0, \sigma_k^2)$, with $\sigma_k^2 > 0$.

(A3) For $k = 1, 2$, $\hat{\sigma}_k^2 \xrightarrow{p} \sigma_k^2$.

Proposition 4.2. *Under (A1)–(A3),*

$$\liminf_{n\to\infty} \inf_{\omega\in[0,1]} \Pr\big(\ell \ge \hat{A}(\omega)\big) \;\ge\; 1 - \alpha/2,$$

where the infimum is over fixed weights. The same bound holds at any weight $\hat{\omega} \in [0, 1]$ computed from the training fold alone, including the estimated oracle weight in (12).

2 Proof

Write $T_n(\omega) = \big(\hat{L}(\omega) - L(\omega)\big)/\sqrt{v_n(\omega)}$, and $\hat{T}_n(\omega)$ for the same ratio with $\hat{v}_n(\omega)$ in place of $v_n(\omega)$. Let $Z_k = \sqrt{n_k}\,(\hat{L}_k - L_k)/\sigma_k$, with distribution function F_k, and

$$\varepsilon_k = \sup_{x\in\mathbb{R}} \big|F_k(x) - \Phi(x)\big|.$$

Z_k, F_k and ε_k all depend on n. By (A2) and Pólya's theorem (convergence to a continuous limit is uniform), $\varepsilon_k \to 0$ for $k = 1, 2$.

Step 1 (reduce to fixed weights). By (A1), $\hat{\omega}$ is fixed given the training fold, and $\hat{A}$ depends only on the evaluation fold. Since $L(\omega)$ is a convex combination of L_1 and L_2, both at most ℓ, we have $L(\omega) \le \ell$. So $L(\omega) \ge \hat{A}(\omega)$ implies $\ell \ge \hat{A}(\omega)$, and $L(\omega) \ge \hat{A}(\omega)$ is the event $\hat{T}_n(\omega) \le z$. Hence

$$\Pr\big(\ell \ge \hat{A}(\hat{\omega})\big) \;\ge\; \inf_{\omega\in[0,1]} \Pr\big(\ell \ge \hat{A}(\omega)\big) \;\ge\; \inf_{\omega\in[0,1]} \Pr\big(\hat{T}_n(\omega) \le z\big).$$

It is enough to bound the right-hand side below by something that tends to $1-\alpha/2$.

Step 2 (normal approximation, uniform in ω). Write $T_n(\omega) = aZ_1 + bZ_2$, with

$$a = \omega\sqrt{s_1/v_n(\omega)}, \qquad b = (1-\omega)\sqrt{s_2/v_n(\omega)}.$$

Then $a, b \ge 0$ and $a^2 + b^2 = 1$. We show

$$\sup_{\omega\in[0,1]} \sup_{x\in\mathbb{R}} \big|\Pr\big(T_n(\omega) \le x\big) - \Phi(x)\big| \;\le\; \varepsilon_1 + \varepsilon_2.$$

If $a = 0$ then $T_n(\omega) = Z_2$ and the bound is immediate. Otherwise, the sources are independent, so conditioning on Z_2 gives

$$\Pr\big(T_n(\omega) \le x\big) = \mathbb{E}\Big[F_1\Big(\frac{x - bZ_2}{a}\Big)\Big].$$

Replacing F_1 by Φ changes this by at most ε_1. The function $y \mapsto \Phi\big((x-by)/a\big)$ is non-increasing with values in $[0,1]$, so replacing Z_2 by $W_2 \sim N(0,1)$ changes the expectation by at most ε_2. (For such a function g, $\mathbb{E}\,g(Y) = \int_0^1 \Pr\big(g(Y) > t\big)\,dt$, and each event $\{g(Y) > t\}$ is a half-line in Y.) Finally, for independent $W_1, W_2 \sim N(0,1)$,

$$\mathbb{E}\Big[\Phi\Big(\frac{x - bW_2}{a}\Big)\Big] = \Pr\big(aW_1 + bW_2 \le x\big) = \Phi(x), \qquad \text{since } a^2 + b^2 = 1.$$

Step 3 (estimated variances). The variance ratio is a weighted average of the two source-level ratios:

$$\frac{\hat{v}_n(\omega)}{v_n(\omega)} = a^2\,\frac{\hat{\sigma}_1^2}{\sigma_1^2} + b^2\,\frac{\hat{\sigma}_2^2}{\sigma_2^2}, \qquad \text{so} \qquad \left|\frac{\hat{v}_n(\omega)}{v_n(\omega)} - 1\right| \le \delta_n := \max_k \left|\frac{\hat{\sigma}_k^2}{\sigma_k^2} - 1\right|$$

for every ω, and $\delta_n \to 0$ in probability by (A3). Fix $\eta \in (0,1)$. On $\{\delta_n \le \eta\}$ we have $\hat{v}_n(\omega) \ge (1-\eta)\,v_n(\omega)$. Since $\hat{T}_n(\omega) \le z$ is the event $T_n(\omega) \le z\sqrt{\hat{v}_n(\omega)/v_n(\omega)}$ and $z > 0$, on $\{\delta_n \le \eta\}$ the event $T_n(\omega) \le z\sqrt{1-\eta}$ implies $\hat{T}_n(\omega) \le z$. With Step 2, for every ω,

$$\Pr\big(\hat{T}_n(\omega) \le z\big) \;\ge\; \Phi\big(z\sqrt{1-\eta}\big) - \varepsilon_1 - \varepsilon_2 - \Pr(\delta_n > \eta).$$

The right-hand side does not depend on ω and tends to $\Phi(z\sqrt{1-\eta})$. Letting $\eta \downarrow 0$ gives $\liminf \ge \Phi(z) = 1 - \alpha/2$, which with Step 1 proves the proposition. □

3 Remarks

1. **The optimisation plays no role in validity.** The proof never uses the definition of the weight in (12); it needs only that $\hat{\omega}$ is computed from the training fold. The optimisation matters only for width.

2. **Uniformity over distributions.** The bound in Step 3 is explicit, so it is uniform over any class of distributions on which ε_1, ε_2 and δ_n tend to 0 uniformly. This is the usual standard in partial identification (Imbens and Manski, 2004; Stoye, 2009). Because $\hat{\omega}$ is fixed given the training fold, nothing breaks when $L_1 = L_2$.